U-AUB-02 Aubrun Bidefect Formalization

Kieran McShane

June 8, 2026

Chapter 1

Corrected Aubrun bidefect route

This blueprint tracks the formal proof route for ticket U-AUB-02. The goal is to close the corrected Aubrun count theorem used by the expectation pipeline. The current route is not the old tight-fiber route and not the false total-defect route. It is the one-sided plus-defect route, converted to a bidefect count and then absorbed by the shifted-base scalar estimate.

1.1 Bidefect endpoint

Theorem 1 (Conditional gap-three Biane frontier with child primitive absorption)**.** *This is not a no-input proof of U-AUB-02. Fix a Wick length m. Assume the remaining large-gap left and right Biane adjacent-link statements outside the checked gap-three slices, assume the side-tied residual branch collision statements: the primitive-residual branches for nonwrapping first return, wrap-below last return, and wrap-above first return; the internal-subinterval branches for wrap-below and wrap-above; and the nonwrapping off-cycle open-internal branch. In the nonwrapping internal-subinterval branch, the checked first-return refinement removes both the endpoint-touching subcase and the subcase where the internal witness remains in the skipped x-cycle; the checked noncrossing confinement lemma below then traps the whole internal witness block strictly inside the return interval. Also assume the Chapuy recurrence*

$$N_{m,g+1} \leq (2(m+1))^3 N_{m,g}.$$

Then the Aubrun bidefect count bound holds at length m. The phrase "checked endpoint" here means checked implication from these hypotheses. It does not mean that the Biane, side-tied return-interval, or Chapuy hypotheses have been proved. This is the current honest theorem face: it keeps the predecessor-skip origin data, still exposes the shorter-interval alternative as an actual internal subinterval of the side-tied return interval, splits the side disjunction into three theorem-facing origin-tag cases, splits each origin-tag case into primitive-residual versus internal-subinterval branches, exposes the nonwrapping internal-subinterval leaf only in its off-cycle open-internal form, sends that off-cycle witness to a smaller side-aware child interval, splits the child into primitive and internal-subinterval branches, and then absorbs the primitive child branch into the already theorem-facing parent primitive residual leaf. The only remaining smaller child input is the genuine internal-subinterval branch. The loose primitive endpoint is kept only as a conditional diagnostic; a checked no-go theorem below shows that its primitive residual input is false without the nested side data. The right gap-three branch theorems listed as dependencies are checked internal dischargers; they are not hypotheses of the endpoint statement. In particular, *the active endpoint no longer exposes the right special-slice parameters corresponding to $S(b, c)$, $b + 1 \sim_\pi c$, or $\pi(b + 1) = b + 1$, and it no longer exposes the final branch $S(b + 1, d)$. It also no longer exposes the left end-slice branch $S(a + 1, c)$. The checked finite gap-three branch list is closed at this endpoint.*

Proof. The checked gap-two Biane theorems supply the adjacent links when the visible gap has length two. The checked gap-three reductions remove both left slices $c - a = 3, b = a + 1$ and $c - a = 3, b = a + 2$, together with the right slice $d - b = 3, c = b + 2$, from the broad large-gap hypotheses. The left start-adjacent slice $c - a = 3, b = a + 1$ is now fully discharged at this endpoint: the $S(b, c)$ branch is supplied from the right frontier, the fixed-$a + 2$ and $\pi(b, a + 2)$ branches force that same contradiction, and the residual $\neg(a \sim_\pi a + 2) \wedge S(a, a + 2)$ branch is contradicted by tree uniqueness for a copied length-two contour path. In the left end-adjacent slice, the fixed $a + 1$ branch is also closed by comparing the copied length-four path $S(a) \to S(c)$ with the direct path from $a \sim_\pi c$. The remaining branch $S(a+1, c)$ forces $a+1 \sim_\pi b$ by comparing the copied contour step $a + 1 \to b$ with the reversed contour step $b \to c$, and the already checked $a+1 \sim_\pi b$ branch absorber closes it; in the right slice, the $\pi(b, c)$ branch is closed by the immediate contradiction with the separating non-link $a \nsim_\pi b$, and the $S(b, c)$ branch is closed by copying the contour step $c \to d$ across $S(b) = S(c)$ and comparing it with the direct path from $b \sim_\pi d$. The $\pi(b+1, c)$ branch is routed through the shifted crossing $(a, b, b+1, d)$: it gives $a \sim_\pi b+1$, the large-gap right input gives $S(b + 1, d)$. The fixed branch $\pi(b + 1) = b + 1$ is also routed through $S(b+1, d)$: fixedness gives the adjacent repeat $S(b+1, c)$, and denying $S(b+1, d)$ creates a length-four incidence path from $S(b)$ to $S(d)$ that contradicts the direct length-two path from $b \sim_\pi d$. Finally, the $S(b+1, d)$ branch itself is closed: denying $S(c, d)$ forces $b+1 \sim_\pi c$ by path uniqueness, the shifted left adjacent link gives $S(a, b)$, and the resulting four incidences form a forbidden square. These links give noncrossing. The direct skip-hit case is consumed internally by the checked theorem `permCyclicPredecessorSkip_direct_commonCycle`. Thus the theorem-facing return-interval theorem only has to handle the nested non-direct case. The branch-normalized return data is supplied internally, the full-endpoint common-cycle branch is discharged internally, and the raw internal right-repeat branch is first replaced by a subinterval-preserving descended pair of alternatives: an adjacent right repeat or a strictly shorter longer same-π subinterval $[a + r, a + s]$ inside the current return interval. The theorem-facing return-collision input is therefore the side-tied subinterval-descended residual frontier, not the loose primitive five-branch frontier. The separate primitive descent theorem is only a conditional diagnostic unless the predecessor-skip side data is carried through descent. In the nonwrapping off-cycle child interval, the checked reductions either find the common pair immediately or preserve the side data on a smaller child residual; the primitive child part of that residual is absorbed back into the parent primitive residual leaf. Thus the remaining smaller child theorem-facing branch is only the genuine internal-subinterval branch. Together with left-tight transversality this rules out non-fixed predecessor skips. The existing genus-zero and Chapuy route then gives the bidefect count bound. This is a conditional reduction theorem: the marker certifies the reduction from the named hypotheses, not the open large-gap Biane, side-tied subinterval return-interval, or Chapuy inputs themselves. □

Theorem 2 (Off-cycle internal blocks are strictly nested)**.** *In the nonwrapping first-return branch, suppose a and c lie in the same left cycle, the transported cycle partition is noncrossing, and the internal subinterval witness starts at a point $a + r$ which is not in the skipped x-cycle. Then every point of the left cycle containing $a + r$ lies strictly between a and c.*

Proof. The points a and c lie in the skipped x-cycle, while $a + r$ is off that cycle. Thus the block of $a + r$ is distinct from the block of a, c, and it contains a point strictly between a and c. The noncrossing interval-nesting lemma confines the whole block to the closed interval $[a, c]$. If one of its points were equal to a or c, the block would meet the skipped x-cycle, contradicting the off-cycle condition. Therefore the confinement is strict. ☐

Theorem 3 (Off-cycle witnesses give smaller contour intervals)**.** *In the same nonwrapping off-cycle branch, the internal witness interval $[a + r, a + s]$ is itself a strictly smaller same-π return interval inside (a, c). It carries the standard* `leftTightContourIntervalRepeat` *obstruction, and every point of its π-block remains off the skipped x-cycle.*

Proof. The endpoints $a + r$ and $a + s$ come directly from the off-cycle internal residual witness. The inequalities $0 < r < s < c - a$ and $1 < s - r$ make this a strictly smaller nontrivial interval. The previous confinement theorem and the first-return condition show that the whole witness block lies in (a, c) and is disjoint from the skipped x-cycle. Finally the generic left-tight contour-repeat theorem applies to the longer same-π interval $[a + r, a + s]$. ☐

Theorem 4 (Off-cycle witnesses reduce to smaller side-aware residuals)**.** *In the nonwrapping off-cycle branch, the smaller contour-repeat interval either already gives a common $\pi/S\pi$ pair, or leaves a strict residual on a smaller interval. In the latter case, no point of the skipped x-cycle lies inside the child interval, and the child π-block remains inside the parent interval and off the skipped cycle.*

Proof. Apply the smaller-contour theorem, then split the resulting contour repeat by the strict-frontier decomposition. The common branch is immediate. In the residual branch, the first-return condition restricts from the parent interval to the child interval, and the block confinement data is carried unchanged. ☐

Theorem 5 (The smaller residual splits into descended child branches)**.** *The side-aware residual child from the nonwrapping off-cycle branch can be decomposed by the subinterval-preserving descent. Thus the theorem-facing child input is not a raw residual predicate: it is either the side-aware primitive residual branch or the side-aware internal-subinterval branch, with the no-x-cycle and block-off-x data still available.*

Proof. Apply the checked descent from a raw strict residual to the subinterval-preserving descended residual. The definitional branch split of that descended residual gives the primitive child case or the internal subinterval child case, and each case is passed to the corresponding side-aware collision input. ☐

Theorem 6 (The smaller primitive child splits into five local branches)**.** *The primitive child residual in the nonwrapping off-cycle branch admits a checked five-way local split: internal S-repeat, start-attached S-repeat, end-attached S-repeat, fixed predecessor, and adjacent T-repeat. Each case still carries the outer nonwrapping first-return data, the no-x-cycle condition inside the child interval, and the block-off-x invariant. The current endpoint no longer exposes these five cases separately; the following arithmetic bridge absorbs them into the parent primitive leaf.*

Proof. Unfold the primitive residual frontier into its definitional five-way branch split. Each branch is passed to the corresponding side-aware child collision input with the inherited first-return, no-x-cycle, and block-off-x hypotheses unchanged. ☐

Theorem 7 (Strict child primitive residuals are parent primitive residuals)**.** *Let $[a', c']$ be a strict child interval inside a parent interval $[a, c]$. If the child interval carries any primitive residual branch, then the parent interval itself carries a primitive residual branch.*

Proof. The proof is arithmetic. Internal, start-attached, and end-attached child S-repeats become internal parent S-repeats after translating the offsets from a' to a. A fixed predecessor at $c' - 1$ gives the adjacent left repeat $(c' - 1, c')$, again an internal parent S-repeat. A child adjacent T-repeat remains an adjacent T-repeat after the same offset translation. Thus no new child primitive collision theorem is needed beyond the parent primitive residual leaf. $\square$

1.2 Checked local Biane inputs

Theorem 8 (Left gap-two Biane link). *For a crossing $a < b < c < d$ with $c - a = 2$, the left adjacent $(\gamma\pi)$-cycle link $a \sim_{\gamma\pi} b$ follows from left-tightness, the crossing links $a \sim_\pi c$, $b \sim_\pi d$, and the separating non-link $a \nsim_\pi b$.*

Theorem 9 (Right gap-two Biane link). *For a crossing $a < b < c < d$ with $d - b = 2$, the right adjacent $(\gamma\pi)$-cycle link $c \sim_{\gamma\pi} d$ follows from the same left-tight crossing data.*

Theorem 10 (Left gap-three start-adjacent reduction). *In the finite slice $c-a = 3$, $b = a+1$, the contour-repeat split first reduces the left Biane link to four explicit start-slice branches: the residual branch $\neg(a \sim_\pi a + 2)$ together with $S(a, a + 2)$, $S(b, c)$, the fixed-point branch $\pi(a + 2) = a + 2$, and $\pi(b, a + 2)$. Later checked suppliers discharge all four at the endpoint, so this reducer is now historical plumbing for the fully closed left start-adjacent slice.*

Theorem 11 (Left start target impossibility). *In the same finite slice $c - a = 3$, $b = a + 1$, the target $S(a, b)$ is incompatible with the nontrivial π-interval $a \sim_\pi c$. Hence each remaining branch-to-target obligation in this start slice is equivalent to proving that the branch itself is impossible.*

Theorem 12 (Residual left-start branch). *In the slice $c - a = 3$, $b = a + 1$, the branch $\neg(a \sim_\pi a + 2) \wedge S(a, a + 2)$ is impossible. The repeat $S(a, a + 2)$ copies the contour step from $a + 2$ to c into a length-two path from $S(a)$ to $S(c)$; comparison with the direct length-two path from $a \sim_\pi c$ forces $a \sim_\pi a + 2$.*

Theorem 13 (Left gap-three end-adjacent reduction). *In the finite slice $c - a = 3$, $b = a + 2$, the contour-repeat split reduces the left Biane link to two explicit branch implications. The adjacent repeat $S(a + 1, b)$ has been normalized, using left-tightness, to the fixed-point branch $\pi(a + 1) = a + 1$. The other remaining branch is $S(a + 1, c)$. The next two theorems close both branches, so this finite slice is no longer theorem-facing on the sharp endpoint.*

Theorem 14 (Left gap-three fixed $a + 1$ branch). *In the same slice $c - a = 3$, $b = a + 2$, fixedness of $a + 1$ gives the adjacent repeat $S(a + 1, b)$. If the target $S(a, b)$ failed, the contour step $a \to a+1$, this repeat, and the contour step $b \to c$ would form a simple length-four incidence path from $S(a)$ to $S(c)$. Tree uniqueness compares it with the direct length-two path from $a \sim_\pi c$, contradiction.*

Theorem 15 (Left gap-three $S(a + 1, c)$ branch). *In the same slice $c - a = 3$, $b = a + 2$, the branch $S(a+1, c)$ forces $a+1 \sim_\pi b$. The copied contour step $a+1 \to b$ is a simple path from $S(c)$ to $S(b)$, and the reversed contour step $b \to c$ is another simple path with the same endpoints. Tree uniqueness identifies their middle right vertices. The already checked $a + 1 \sim_\pi b$ branch theorem then closes the target.*

Theorem 16 (The $\pi(a, a+1)$ end-adjacent branch). *In the same slice, the branch $\pi(a, a+1)$ is absorbed by the shifted gap-two theorem: $\pi(a, a+1)$ and $\pi(a, c)$ imply $\pi(a+1, c)$, so the crossing $(a + 1, b, c, d)$ gives $S(a + 1, b)$, which is one of the remaining explicit branches.*

Theorem 17 (The $\pi(a+1,b)$ end-adjacent branch)**.** *In the same slice, the branch $\pi(a+1,b)$ is absorbed by the already exposed start-adjacent branch on the shifted crossing $(a,a+1,c,d)$. That branch gives the adjacent repeat $S(a,a+1)$, which contradicts the nontrivial same-π interval $a \sim_\pi c$.*

Theorem 18 (Right gap-three end-adjacent reduction)**.** *In the finite slice $d-b=3$, $c=b+2$, the contour-repeat split and the shifted gap-two branch remove the $\pi(b,b+1)$ case, while transitivity with $a \sim_\pi c$ removes the $\pi(b,c)$ case. The remaining branches at this reducer level are $S(b,c)$, $\pi(b+1)=b+1$, $S(b+1,d)$, and $\pi(b+1,c)$. The adjacent repeat $S(b+1,c)$ has been normalized, using left-tightness, to the fixed-point branch. The separate theorem below closes the $S(b,c)$ branch at the current endpoint, and the next theorem routes the $\pi(b+1,c)$ branch through $S(b+1,d)$. The fixed-branch theorem after that routes $\pi(b+1)=b+1$ through the same $S(b+1,d)$ branch, and the final theorem below closes $S(b+1,d)$ itself. Thus the active endpoint has no theorem-facing right branch in this checked slice.*

Theorem 19 (Right gap-three $S(b,c)$ branch)**.** *In the finite slice $d-b=3$, $c=b+2$, the branch $S(b,c)$ forces the target $S(c,d)$. If $S(c,d)$ failed, the contour step from c to d, copied across $S(b)=S(c)$, would be a simple path from $S(b)$ to $S(d)$. Tree uniqueness compares it with the direct path from $b \sim_\pi d$, forcing $b \sim_\pi c$, which contradicts the crossing link $a \sim_\pi c$ and the separating non-link $a \nsim_\pi b$.*

Theorem 20 (Right gap-three $\pi(b+1,c)$ branch)**.** *In the finite slice $d-b=3$, $c=b+2$, the branch $b+1 \sim_\pi c$ gives $a \sim_\pi b+1$ by transitivity with the crossing link $a \sim_\pi c$. Applied to the shifted crossing $(a,b,b+1,d)$, the large-gap right input supplies $S(b+1,d)$, because this shifted crossing is not the special $c=b+2$ slice. The final right-branch theorem below consumes $S(b+1,d)$, so $b+1 \sim_\pi c$ is no longer theorem-facing.*

Theorem 21 (Right gap-three fixed $b+1$ branch)**.** *In the finite slice $d-b=3$, $c=b+2$, the fixed branch $\pi(b+1)=b+1$ gives the adjacent repeat $S(b+1,c)$. If $S(b+1,d)$ failed, the contour step $b \to b+1$, followed by the copied contour step $c \to d$, would form a simple length-four incidence path from $S(b)$ to $S(d)$. Tree uniqueness compares it with the direct length-two path coming from $b \sim_\pi d$, contradiction. Hence the fixed branch is no longer theorem-facing at the active endpoint; it is fed to the final $S(b+1,d)$ branch theorem.*

Theorem 22 (Right gap-three $S(b+1,d)$ branch)**.** *In the finite slice $d-b=3$, $c=b+2$, the branch $S(b+1,d)$ forces the target $S(c,d)$, provided the shifted left adjacent link on $(a,b,b+1,d)$ is available. If $S(c,d)$ failed, uniqueness of the two simple incidence paths from $S(c)$ to $S(d)$ would force $b+1 \sim_\pi c$. Then $a \sim_\pi c$ gives $a \sim_\pi b+1$, the shifted left link gives $S(a,b)$, and the links $a \sim_\pi b+1$, $b \sim_\pi d$, $S(a,b)$, and $S(b+1,d)$ form a forbidden incidence square.*

1.3 Skip frontier

Theorem 23 (Minimal frontier extracted from a non-fixed skip)**.** *Any non-fixed predecessor skip in a left-tight permutation produces a minimal long same-π interval. Its contour frontier has five possible alternatives: an adjacent right repeat, an internal left repeat, a proper start-touching left repeat, a proper end-touching left repeat, or a fixed predecessor at the right endpoint.*

Theorem 24 (Tied contour interval extracted from a non-fixed skip)**.** *Any non-fixed predecessor skip in a left-tight permutation produces a long repeated contour interval whose endpoints remember the skip witness. In the nonwrapping case the interval is exactly $[(\pi x).\mathrm{val}, x.\mathrm{val}]$; in the wrapping-below case it ends at $(\pi x).\mathrm{val}$ and starts below x; in the wrapping-above case it starts at $x.\mathrm{val}$ and ends above $(\pi x).\mathrm{val}$.*

Theorem 25 (First return in the nonwrapping nested skip branch)**.** *In the nonwrapping branch, if $\gamma(\pi x)$ is not in the π-cycle of x, then the first same-π return after $(\pi x).\mathrm{val}$ is separated by a genuine gap. The resulting interval starts at $(\pi x).\mathrm{val}$, has no earlier same-cycle return to x, and carries the repeated-contour obstruction.*

Theorem 26 (First return in the wrap-above nested skip branch)**.** *In the wrap-above branch, the first same-π return above $(\pi x).\mathrm{val}$ is also separated from πx by a genuine gap unless the direct-hit case occurs. The corresponding long interval starts at $x.\mathrm{val}$ and carries the repeated-contour obstruction.*

Theorem 27 (Branch-normalized nested skip return)**.** *In a genuinely nested non-fixed predecessor skip, the canonical return interval is now available in every branch: first return in the nonwrapping branch, last return in the wrap-below branch, and first return in the wrap-above branch. Each interval has a genuine gap and carries the repeated-contour obstruction supplied by left-tightness.*

Theorem 28 (Strict return-interval frontier)**.** *A branch-normalized repeated-contour interval can be sharpened to a strict return frontier. The full endpoint repeat is already a common-cycle collision. The start-adjacent left repeat is impossible by left-tightness, and the end-adjacent left repeat is exactly the fixed-predecessor branch. The theorem-facing strict alternatives are therefore internal left repeat, proper start repeat, proper end repeat, fixed predecessor, or internal right repeat.*

Theorem 29 (Residual return-interval frontier)**.** *The strict return frontier splits into an already-solved common-cycle branch and a residual frontier. The active endpoint consumes only the residual frontier; the common-cycle branch is discharged internally.*

Theorem 30 (Descended residual return frontier)**.** *The residual internal right-repeat branch is no longer left raw. A checked descent replaces it by either an adjacent same-π repeat or a strictly shorter longer same-π interval. This is a narrowing of the nested collision input, not a proof of that collision input.*

Theorem 31 (Subinterval-descended residual return frontier)**.** *The residual internal right-repeat branch is narrowed without losing the side-tied return-interval geometry. Adjacent internal right repeats become the adjacent primitive branch; non-adjacent internal right repeats are kept as an actual shorter subinterval $[a + r, a + s]$ of the current return interval. Thus the theorem-facing recursive branch is no longer an arbitrary shorter same-π interval.*

Theorem 32 (Loose primitive residual no-go)**.** *The primitive residual collision statement is false if stated without the nested predecessor-skip origin data. Concretely, at $m = 2$, the inverse long cycle is left-tight and noncrossing, the interval $(0, 2)$ satisfies the primitive residual predicate through the adjacent-π branch, but $\gamma\pi = 1$, so there is no nontrivial common π- and $\gamma\pi$-cycle pair.*

Theorem 33 (Primitive residual descent)**.** *The strictly shorter same-π interval alternative in the descended residual frontier is closed by well-founded descent on interval length. It is enough for the theorem-facing collision input to solve the five primitive residual branches: internal left repeat, proper start repeat, proper end repeat, fixed predecessor, and adjacent right repeat. This theorem is not the current endpoint bridge by itself, because its primitive input does not carry the predecessor-skip side data.*

1.4 Open leaves

Theorem 34 (Large-gap left Biane link)**.** *For every genuine crossing with* $2 < c{-}a$ *and* $c{-}a \neq 3$*, prove the left adjacent link* $a \sim_{\gamma\pi} b$*. This is still open.*

Theorem 35 (Large-gap right Biane link)**.** *For every genuine crossing with* $2 < d - b$ *outside the checked slice* $d - b = 3,\ c = b + 2$*, prove the right adjacent link* $c \sim_{\gamma\pi} d$*. This is still open.*

Theorem 36 (Nested return-interval branch collision)**.** *Prove the side-tied residual branch collision statements consumed by the current endpoint, or replace them by a side-descending primitive theorem. These are the primitive-residual branches for nonwrapping first return, wrap-below last return, and wrap-above first return; the internal-subinterval branches for wrap-below and wrap-above; and the nonwrapping off-cycle open-internal branch. In the nonwrapping internal-subinterval case, the endpoint-touching start* $r = 0$ *and the* x*-cycle internal-witness subcase are already discharged by the first-return condition, so the remaining leaf is the off-cycle open-internal case. The theorem must carry the origin tag or carry equivalent context through interval descent. The raw internal right-repeat branch has been replaced by adjacent-or-subinterval descent, and the recursive branch now remembers the internal offsets* $[a + r, a + s]$ *inside the side-tied return interval. What remains is not the loose primitive frontier by itself: Lean gives a counterexample to that over-broad statement. The remaining theorem must use the nested side/context data and a genuine noncrossing/incidence-tree argument.*

Theorem 37 (Chapuy recurrence)**.** *Prove the slicing recurrence*

$$N_{m,g+1} \leq (2(m+1))^3 N_{m,g}.$$

This is still open.